EFY TESTED
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IoT-Based Distribution Transformer CONDITION MONITORING SYSTEM

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The proposed IoT-based distribution transformer condition monitoring system enables real-time monitoring of distribution transformers, identifying such deviations as overload conditions and overheating. Through IoT technology, essential operational data is collected and displayed locally on an LCD interface. Additionally, this data is transmitted remotely to cloud platforms like Adafruit.io for analysis, allowing for proactive maintenance. The system's architecture incorporates sensors interfaced with a microcontroller, ensuring seamless data acquisition and transmission. This system empowers utilities to optimise power distribution networks and enhance operational reliability cost-effectively.

In the system design, the ACS712 current sensor detects the transformer's current and identifies overload conditions, while the DHT humidity and temperature sensor monitors the temperature of both the transformer and its surroundings. The ultrasonic sensor tracks the transformer's oil level. The Adafruit IO dashboard enables data monitoring on an IoT platform. The ESP32 board connects the system to a Wi-Fi network, sending data to the IoT dashboard. The authors' prototype is shown in Fig. 1, and

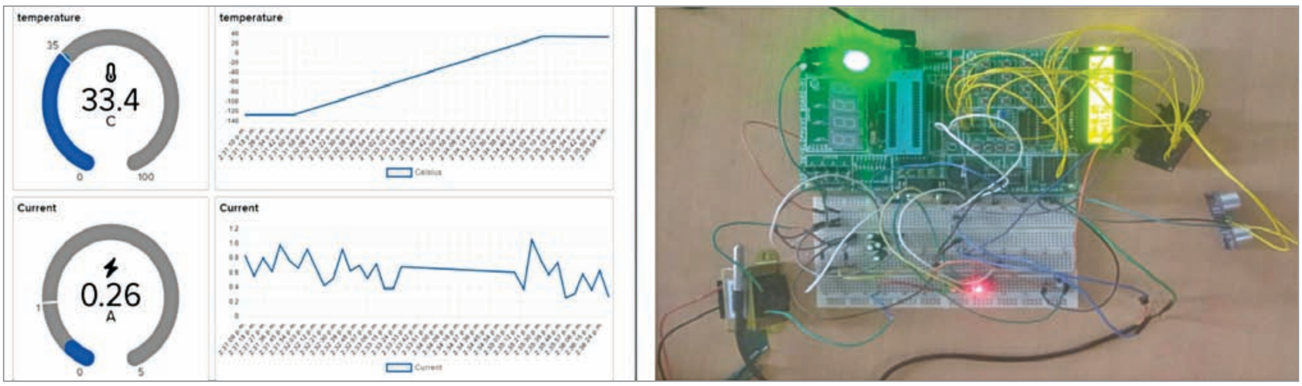

Fig. 1: Authors' prototype

```
#define D_34_A16 34 // ADC 16
#define D_35_A17 35 // ADC 17

#include <WiFi.h>
#include "Adafruit_MQTT.h"
#include "Adafruit_MQTT_Client.h"

#define WLAN_SSID  "ABCDEFGHI"
#define WLAN_PASS "12345678"
#define AIO_SERVER "io.adafruit.com"
#define AIO_SERVERPORT 1883
#define AIO_USERNAME "Abcdefghi_123"
#define AIO_KEY "aio_yUFq50H1pi3aSqrlZai0brRJm"

WiFiClient client;
Adafruit_MQTT_Client mqtt(&client, AIO_SERVER, AIO_SERVERPORT, AIO_USERNAME, AIO_KEY);
Adafruit_MQTT_Publish Feed1 = Adafruit_MQTT_Publish(&mqtt, AIO_USERNAME "/feeds/Current");
Adafruit_MQTT_Publish Feed2 = Adafruit_MQTT_Publish(&mqtt, AIO_USERNAME "/feeds/voltage");
Adafruit_MQTT_Publish Feed3 = Adafruit_MQTT_Publish(&mqtt, AIO_USERNAME "/feeds/Celsius");
Adafruit_MQTT_Publish Feed4 = Adafruit_MQTT_Publish(&mqtt, AIO_USERNAME "/feeds/level");
```

Fig. 2: Configuring the Adafruit MQTT API in code

BILL OF MATERIALS

Components	Quantity
ESP32 board	1
Transformer	1
Current sensor ACS712	1
Humidity and temperature sensor DHT11	1
Ultrasonic sensor HC-SR04	1
Wires and jumpers	As required
16x2 LCD display module	1

Note. *The LCD display is optional for local data monitoring. If it is not connected, data will still be displayed remotely on the IoT dashboard.*